

SIMFEROPOL, Ukraine

WMO: 339460

Lat: 45.0403N

Lon: 33.9672E

Elev: 594

StdP: 14.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	15.2	4.8	7.2	11.5	9.3	9.0	17.0	35.3	24.6	32.2	26.5	13.6	50	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	92.8	68.8	89.6	68.2	86.4	67.1	72.5	83.8	70.9	82.4	69.5	81.1	13.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.4	110.4	76.6	67.7	103.9	75.5	66.0	97.9	74.1	36.5	83.1	35.2	82.3	34.0	81.4	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	24.6	21.6	3.5	97.2	7.2	2.7	-1.7	99.1	-5.9	100.7	-10.0	102.2	-15.2	104.2
			3.0	74.8	7.0	1.6	-2.0	76.0	-6.1	76.9	-10.0	77.8	-15.1	79.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	32.8	34.6	41.0	51.1	60.8	68.7	74.0	73.7	64.2	53.5	44.8	36.7
	DBStd	16.42	9.82	10.17	8.12	7.10	6.47	5.38	4.76	5.51	6.44	7.95	8.61	9.59
	HDD50	2017	535	437	296	68	4	0	0	0	1	55	200	421
	HDD65	5137	998	852	743	419	161	24	1	4	89	365	605	876
	CDD50	3149	2	5	18	101	339	561	744	734	428	162	45	10
	CDD65	795	0	0	0	2	31	134	280	274	66	8	0	0
	CDH74	7562	0	0	1	35	309	1179	2705	2716	551	62	4	0
	CDH80	2817	0	0	0	3	53	357	1098	1163	133	10	0	0
Wind	WSAvg	10.2	11.2	12.0	11.4	10.7	9.1	8.9	9.1	9.0	9.5	9.8	10.3	11.0
Precipitation	PrecAvg	19.5	1.7	1.3	1.4	1.3	1.5	2.1	2.0	1.8	1.5	1.5	1.7	1.9
	PrecMax	30.8	4.3	3.4	3.7	3.7	5.4	5.1	6.3	11.5	3.8	4.8	4.4	4.4
	PrecMin	12.2	0.2	0.2	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.1	0.1	0.1
	PrecStd	4.1	1.1	0.7	0.8	0.9	1.1	1.2	1.5	1.9	0.9	1.1	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	62.9	68.3	78.7	85.2	91.8	96.4	97.1	89.1	80.6	72.1	62.7
		MCWB	48.6	50.1	51.4	58.7	64.7	68.4	69.7	69.0	66.2	61.9	57.2	51.5
	2%	DB	53.2	56.7	62.9	72.8	80.8	87.7	92.6	93.2	83.9	74.8	66.2	57.0
		MCWB	47.0	48.4	49.9	56.1	63.6	67.2	69.1	68.9	64.6	60.3	54.9	50.0
	5%	DB	49.1	52.2	58.5	68.3	77.3	84.1	89.5	90.0	80.3	70.0	61.2	53.5
		MCWB	45.0	46.3	47.9	54.4	62.3	66.4	68.7	68.1	63.2	58.4	53.2	48.7
	10%	DB	45.9	48.4	53.7	64.3	73.7	80.8	86.2	86.6	76.7	65.7	56.9	49.7
		MCWB	42.9	44.0	46.1	52.4	61.0	65.3	67.6	67.1	62.3	57.0	51.2	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	52.2	53.6	60.7	68.4	72.9	74.5	75.0	70.4	65.0	59.4	54.2
		MCDB	55.5	59.3	64.3	74.9	78.8	85.3	85.6	86.7	79.4	74.8	66.8	59.6
	2%	WB	47.7	49.5	50.9	57.7	65.9	70.2	72.4	72.6	67.5	62.3	56.5	51.3
		MCDB	51.7	55.2	60.1	70.1	76.8	80.6	83.4	84.2	77.2	70.0	64.0	55.6
	5%	WB	45.4	47.0	48.9	55.4	63.9	68.5	71.0	70.9	65.6	60.3	54.1	49.2
		MCDB	48.7	51.7	56.7	65.9	73.8	78.8	82.3	83.0	75.4	67.3	60.1	53.0
	10%	WB	42.8	44.1	46.8	53.4	62.0	67.1	69.7	69.3	63.9	58.0	51.8	46.5
		MCDB	45.3	47.7	52.9	62.7	71.5	77.1	81.4	81.3	73.0	64.6	56.3	49.6
Mean Daily Temperature Range	5% DB	MDBR	11.6	13.2	15.9	19.4	20.3	19.9	21.4	22.0	19.8	16.8	14.2	11.4
		MCDBR	16.6	19.1	23.6	25.6	25.3	24.7	25.8	26.3	25.1	22.8	20.5	15.9
		MCWBR	11.8	12.6	13.7	13.5	12.2	10.1	9.6	9.8	10.6	11.5	12.1	11.1
	5% WB	MCDBR	15.5	17.1	20.8	22.9	22.8	20.9	22.3	23.0	21.2	20.3	18.1	14.5
		MCWBR	11.7	11.9	12.9	12.9	12.0	9.9	9.5	10.3	10.7	11.5	11.7	10.9
Clear-Sky Solar Irradiance	taub	0.335	0.360	0.387	0.426	0.404	0.405	0.424	0.425	0.399	0.366	0.347	0.333	
	taud	2.355	2.286	2.234	2.148	2.299	2.329	2.290	2.273	2.319	2.385	2.379	2.362	
	Ebn at Noon	242	257	266	264	275	274	267	262	259	252	235	229	
	Edh at Noon	25	32	38	45	40	39	41	40	35	29	24	23	
All-Sky Solar Radiation	RadAvg	363	631	1046	1583	1990	2133	2153	1892	1382	856	466	314	
	RadStd	46	77	96	142	142	134	98	108	115	82	50	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.12	N/A	N/A	N/A	+0.97	+1.01	-184	-248	+209	+115
Regional (5 neighbors)		+1.21	N/A	N/A	+1.55	N/A	N/A	-214	-301	+254	+157

Nomenclature: See separate page